2/3/26

Normalization

Normalization is a technique applied as a part of data preperation, to change the value of numeric column without disorting difference in range of values / losing information.

Types:

- → Min-max scaling ⟹ Most commonly used (90% of time this is used)
- → Mean normalization
- → Max absolute scaling
- → Robust scaling.

Min-max scaling

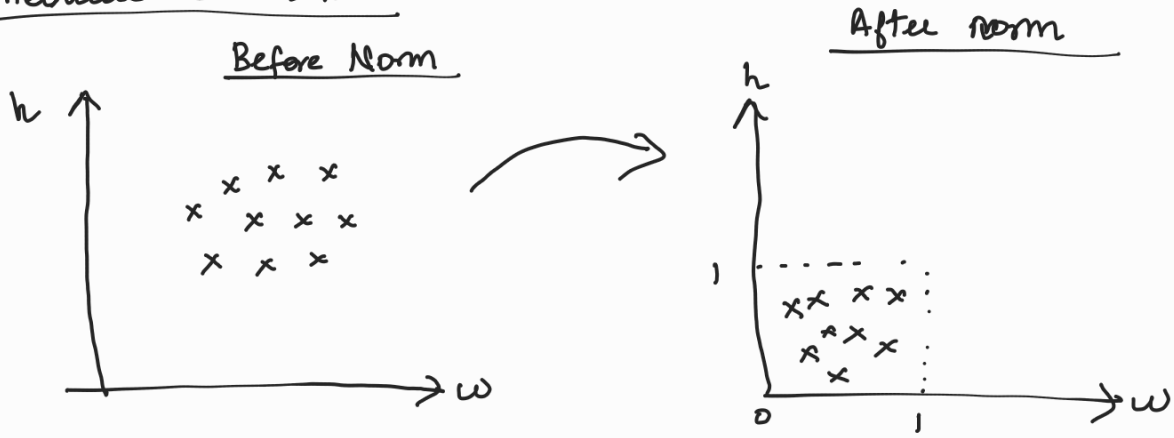
Weight (x_i)		(x_i')
130		1
67	Normalization →	0...
81		0...
61		1
50		0..
32		
⋮		

100 values

$$x_i' = \frac{x_i - x_{min}}{x_{max} - x_{min}}$$

If we see the result after normalization, the values will be b/w the range [0, 1]

Geometrical intuition



Mean normalization :-

Weight (x_i)

200
100
80
95
⋮

$\xrightarrow[\text{normalisation}]{\text{mean}}$

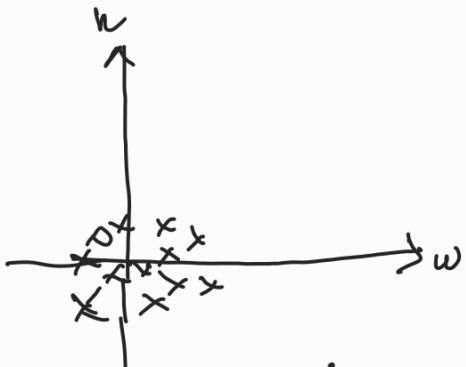
$$x_i' = \frac{x_i - x_{mean}}{x_{max} - x_{min}}$$

$x_i - x_{mean}$ → mean centering like standardisation

$x_{max} - x_{min}$ → Gives the range.

$x_i < \text{mean} \rightarrow$ −ve value

$x_i > \text{mean} \rightarrow$ +ve value.



It is used when the dataset is Centered

Max absolute scaling

Weight (x_i)

200
100
150
⋮

$\xrightarrow[\text{Absolute Scaling}]{\text{max}} x_i'$

$$x_i' = \frac{x_i}{|x_{max}|}$$

This technique is used when the dataset is sparse [0's].

Robust scaling

$$x_i' = \frac{x_i - x_{median}}{IQR\{75^{th}\text{ percentile} - 25^{th}\text{ percentile}\}}$$

IQR → Inter Quatile Range.

This technique is used when the dataset is having outliers.

Normalization Vs standardization

Most problem can be solved by standardisation.

1. Ast if the model requires scaling? (Because some model uses decision tree algorithm which doesn't require scaling)

2. If Normalisation then in most cases it can be solved using min-max scaling.

3. Know the dataset clearly if it has outliers / sparse / centered. And know which algorithm we are using based on that we can select b/w normalisation & standardization.